```
@relation iris-weka.filters.unsupervised.attribute.Remove-R1-2

@attribute petallength numeric
@attribute petalwidth numeric
@attribute class {Iris-setosa,Iris-versicolor,Iris-virginica}

@data
1.4,0.2,Iris-setosa
1.4,0.2,Iris-setosa
1.3,0.2,Iris-setosa
1.5,0.2,Iris-setosa
1.4,0.2,Iris-setosa
1.7,0.4,Iris-setosa
1.4,0.3,Iris-setosa
1.5,0.2,Iris-setosa
1.4,0.2,Iris-setosa
1.5,0.1,Iris-setosa
1.5,0.2,Iris-setosa
1.6,0.2,Iris-setosa
```

Figure 1: ARFF file format

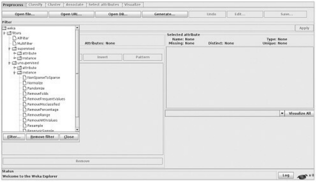

Figure 2: Weka - Explorer (preprocess)

Classify: The next option in Weka Explorer is the Classifier, which is a model for predicting nominal or numeric quantities and includes various machine learning techniques like decision trees and lists, instance-based classifiers, support vector machines, multi-layer perceptrons, logistic regression, Bayes' networks, etc. Figure 3 shows an example of a decision tree using the J4.8 algorithm to classify the IRIS data set into different types of IRIS plants, depending upon some attributes' information like sepal length and width, petal length and width, etc. It provides an option to use a training set and supplied test sets from existing files, as well as cross validate or split the data into training and testing data based on the percentage provided. The classifier output gives a detailed summary of correctly/incorrectly classified instances, mean absolute error, root mean square error, etc.

Cluster: The Cluster panel is similar to the Classify panel. Many techniques like k-Means, EM, Cobweb, X-means and Farthest First are implemented. The output in this tab contains the confusion matrix, which shows how many errors there would be if the clusters were used instead of the true class.

Associate: To find the association on the given set of input data, 'Associate' can be used. It contains an implementation of the Apriori algorithm for learning association rules. These algorithms can identify statistical dependencies between groups of attributes, and compute

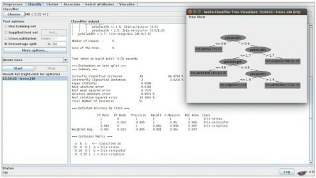

Figure 3: Classification using Weka

all the rules that have a given minimum support as well as exceed a given confidence level. Here, association means how one set of attributes determines another set of attributes and after defining minimum support, it shows only those rules that contain the set of items out of the total transaction. Confidence indicates the number of times the condition has been found true.

Select Attributes: This tab can be used to identify the important attributes. It has two parts — one is to select an attribute using search methods like best-first, forward selection, random, exhaustive, genetic algorithm and ranking, while the other is an evaluation method like correlation-based, wrapper, information gain, chi-squared, etc.

Visualize: This tab can be used to visualise the result. It displays a scatter plot for every attribute.

The components of Experimenter

The Experimenter option available in Weka enables the user to perform some experiments on the data set by choosing different algorithms and analysing the output. It has the following components.

Setup: The first one is to set up the data sets, algorithms output destination, etc. Figure 4 shows an example of comparing the J4.8 decision tree with ZeroR on the IRIS data set. We can add more data sets and compare the outcome using more algorithms, if required.

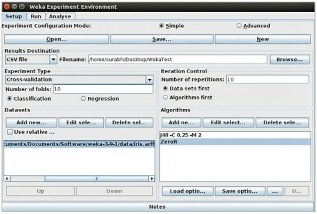

Figure 4: Weka Experimental environment